GIRIDHARAN S

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Summary

Computer Science undergraduate at VIT Vellore (CGPA 9.19) developing robotics, AI and software systems. Seeking internships in robotics, AI/ML or software engineering.

Skills

Languages: Python, C, C++, Java, JavaScript, SQL
AI/ML & Vision: scikit-learn, OpenCV, neuro-symbolic AI
Robotics: ROS 2, RViz, Gazebo, LiDAR, URDF, Nav2 (learning)
Web & Databases: React, Vite, HTML, CSS, FastAPI, Flask, PostgreSQL, PL/pgSQL, SQLite
Tools: Git, GitHub, VS Code, Jupyter Notebook, pytest, Mininet, ESP32
Concepts: Data Structures & Algorithms, OOP, DBMS, Compiler Design

Experience

Junior Software Engineer **Jul 2026 – Present**
Team Artemis, VIT Vellore
- Mapped the Team Artemis lab via physical LiDAR and a simulated Gazebo maze using ROS 2, Hexapod and TurtleBot.
- Executed 2 navigation tests to predefined goals on saved maps while exploring a modified Nav2 approach.
- Assisted juniors with RoboReach (Techfest, IIT Bombay) and supported EYIC and Technocian finals preparation.

AI/ML Intern **May 2025 – Jun 2025**
IITM Pravartak (IIT Madras), Chennai (Hybrid)
- Validated ∼800 Hindi speech-to-text test cases from Sansad TV broadcasts against ground-truth transcripts.
- Evaluated transcripts for accuracy, consistency and edge cases, logging findings to guide further iterations.

Projects

Knowledge-Augmented Neuro-Symbolic Travel Planner | Python, Flask, SQLite, scikit-learn, pytest | GitHub
- Developed an AI agent that converts natural-language trip requests into constrained flight × hotel plans.
- Designed a parse-only language layer feeding a symbolic core for constraints, ranking, risk tiers and validation.
- Implemented plan-aware, value-of-information elicitation: 6 instead of 9 first-trip questions at 100% oracle match.
- Integrated SQLite preference memory and a logistic-regression ranker; validated with 158 automated tests.

GreenPlace – SDN Controller Placement (Group Project) | Python, ILP, NSGA-II, Mininet, Ryu | GitHub
- Formulated energy-aware SDN controller placement under hard delay limits and switch-migration costs.
- Optimized static placement with an exact ILP baseline and NSGA-II across 46 Internet Topology Zoo networks.
- Engineered a runtime manager for dynamic traffic; Mininet/Ryu emulation verified 212 switch migrations.

CineTrace – Explainable and Adaptive Movie Recommender | React, FastAPI, PostgreSQL, pytest
- Architected an 18-table PostgreSQL schema linking each explanation by foreign key to its supporting rows.
- Developed trigger-based, append-only rating history and a time-travel SQL verifier: 0 of 4,294 claims invalid.
- Enforced integrity with deferred and immutability triggers; applied Maximal Marginal Relevance (MMR) re-ranking.
- Tested constraints, triggers, transaction rollback and the API with 40 automated tests.

Mini Compiler for a Custom Language (Team of 3) | Python, unittest, React
- Engineered a seven-phase compiler, from lexer to cycle-counting VM, supporting if/while and functions.
- Devised a cost-gated optimizer that applies a rewrite only if cycles drop: 29.8% fewer VM cycles than standard passes.
- Implemented graph-coloring register allocation and a learned cost model; verified with 81 unit tests and differential testing.

Certifications & Achievements

- **Robotic Fundamentals and Introduction to ROS:** VIT Value Added Course, Jan–Apr 2026.
- **Machine Learning & AI Fundamentals:** TEACHNOOK training program, Oct–Dec 2024.
- **Problem Solving:** Solved 55+ problems on LeetCode (arrays, strings, DP, graphs).

Education

Vellore Institute of Technology (VIT), Vellore **Aug 2024 – May 2028**
B.Tech in Computer Science *CGPA: 9.19 / 10 (till date)*

Samsidh International School **2022 – 2024**
Class XII *85%*

Chinmaya Vidyalaya **2015 – 2022**
Class X *88%*